

AWS Task-2

Task Description:

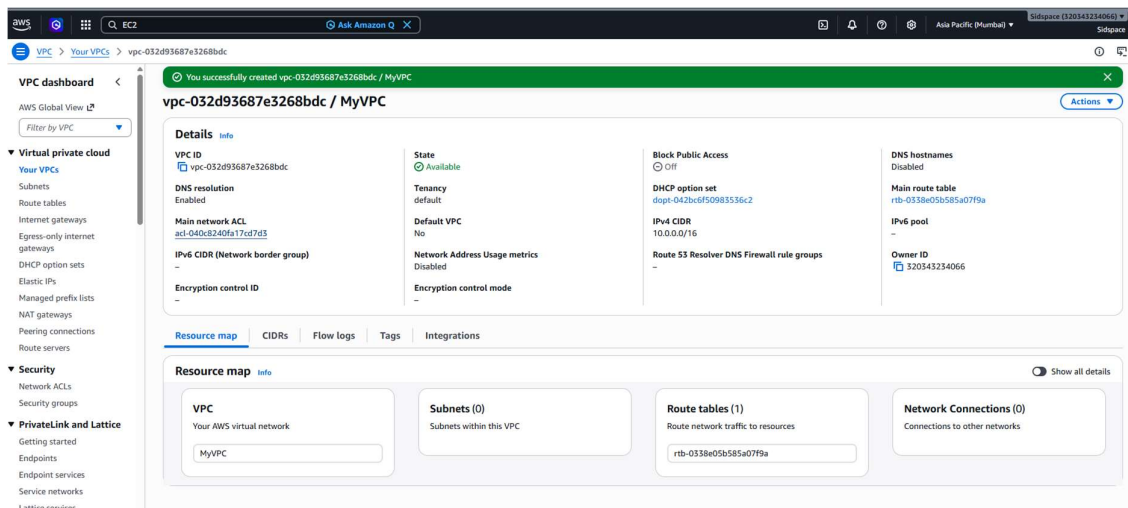
Set up a VPC with an Internet gateway, create a public subnet with 256 IP addresses, a private subnet with 256 IP addresses, make a route table connecting the Internet gateway and the subnets, and launch a Linux EC2 instance by using the above VPC and public subnet.

Solution :

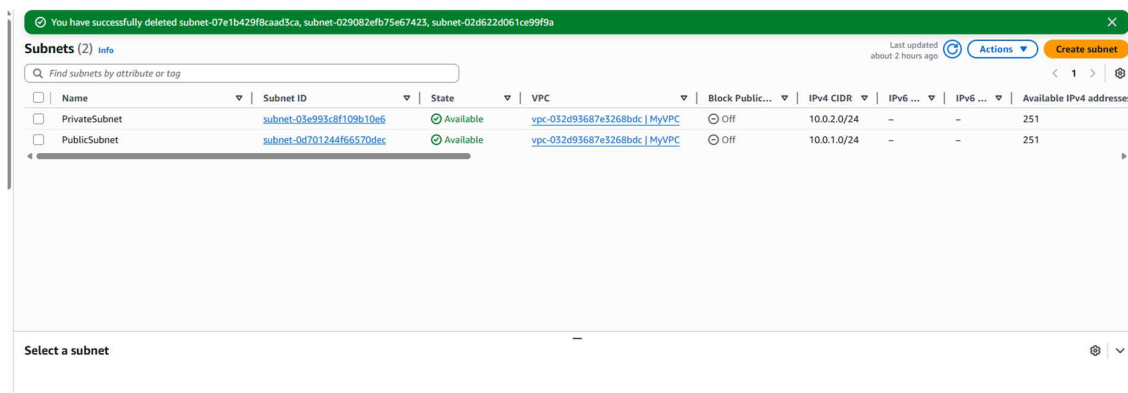
Created a VPC

Name: MyVPC

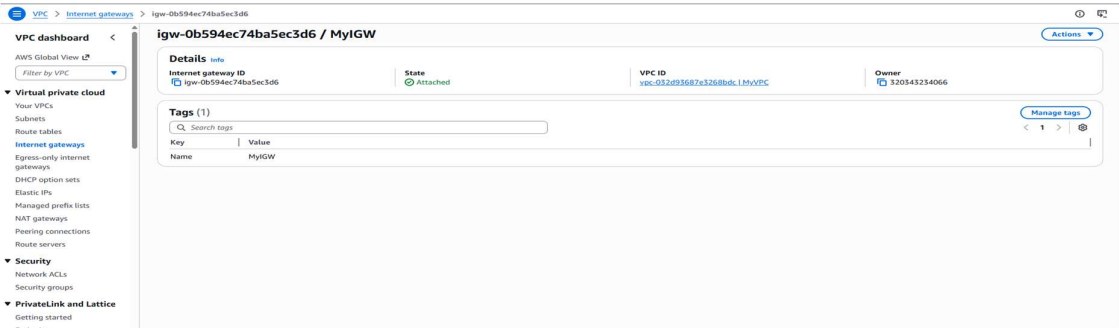
IPv4 CIDR: 10.0.0.0/16



Created Subnets : Public and private CIDR: 10.0.1.0/24



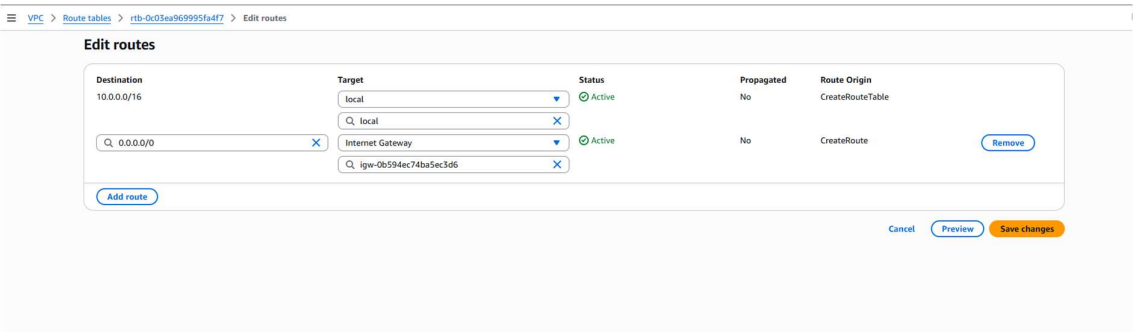
Created Internet Gateway attached to VPC



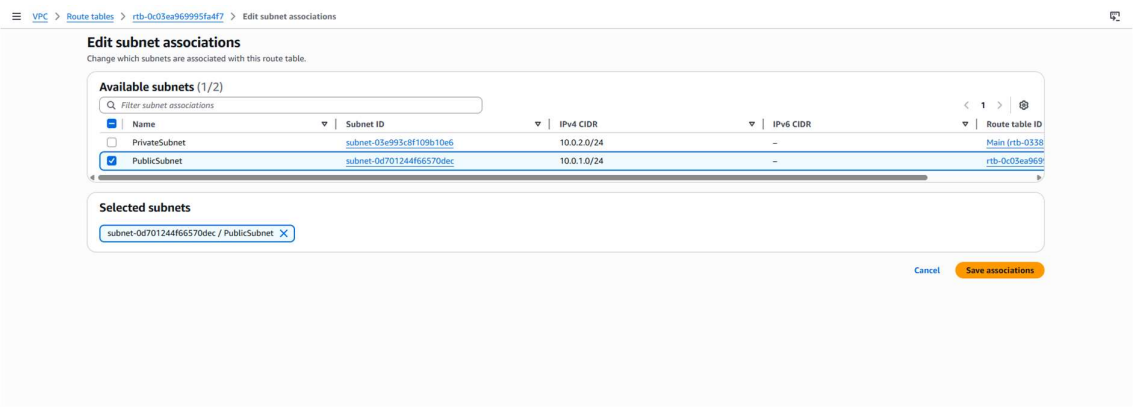
Create Route Table

Destination: 0.0.0.0/0

Target: Internet Gateway



Subnet associations : publicsubnet



Enabled Auto Public IP in Public subnet

Edit subnet settings [Info](#)

Subnet

Subnet ID
subnet-0d701244f66570dec

Name
PublicSubnet

Auto-assign IP settings [Info](#)

Enable AWS to automatically assign a public IPv4 or IPv6 address to a new primary network interface for an instance in this subnet.

☒ Enable auto-assign public IPv4 address [Info](#)

☐ Enable auto-assign customer-owned IPv4 address [Info](#)
Option disabled because no customer owned pools found.

Resource-based name (RBN) settings [Info](#)

Specify the hostname type for EC2 instances in this subnet and optional RBN DNS query settings.

☐ Enable resource name DNS A record on launch [Info](#)

☐ Enable resource name DNS AAAA record on launch [Info](#)

Hostname type [Info](#)

☐ Resource name

☒ IP name

DNS64 settings

Enable DNS64 to allow IPv6-only services in Amazon VPC to communicate with IPv4-only services and networks.

☐ Enable DNS64 [Info](#)

[Cancel](#) [Save](#)

Launched EC2 Instance : chose keypair .pem and network settings to MyVPC public subnet and allowed SSH port 22 my IP and HTTP port 80 to anywhere

Instance summary for i-05168628333c715c9 (MyLinuxInstance) [Info](#)

[Connect](#) [Instance state](#) [Actions](#)

Updated less than a minute ago

Instance ID
i-05168628333c715c9

IPv6 address
-

Hostname type
IP name: ip-10-0-1-187.ap-south-1.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
13.234.239.60 [Public IP]

IAM role
-

IMDSv2
Required

Operator
-

Public IPv4 address
13.234.239.60 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-0-1-187.ap-south-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-032d93687e3268bdc (MyVPC)

Subnet ID
subnet-0d701244f66570dec (PublicSubnet)

Instance ARN
arn:aws:ec2:ap-south-1:320343234066:instance/i-05168628333c715c9

Private IPv4 addresses
10.0.1.187

Public DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Security details

IAM role
-

Security groups
sg-0496bd4f635676327 (launch-wizard-2)

Owner ID
320343234066

Launch time
Sun May 03 2026 13:44:27 GMT+0530 (India Standard Time)

sg-0496bd4f635676327 - launch-wizard-2 [Actions](#)

Details

Security group name
launch-wizard-2

Security group ID
sg-0496bd4f635676327

Description
launch-wizard-2 created 2026-05-03T08:03:44.329Z

VPC ID
vpc-032d93687e3268bdc

Owner
320343234066

Inbound rules count
2 Permission entries

Outbound rules count
1 Permission entry

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2)

[Manage tags](#) [Edit inbound rules](#)

Q Search

☐ Name

Security group rule ID

IP version

Type

Protocol

Port range

Source

Description

☐ -

sg-07590156b272d4b57

IPv4

HTTP

TCP

80

0.0.0.0/0

-

☐ -

sg-03c77d96eddbef2b3

IPv4

SSH

TCP

22

0.0.0.0/0

-

Connected to EC2 from local machine using .pem key and SSH connection built using public ip of EC2

```
Siddharth@LAPTOP-L582J1AR MINGW64 ~/Downloads
$ chmod 400 taskkeypair.pem

Siddharth@LAPTOP-L582J1AR MINGW64 ~/Downloads
$ ssh -i taskkeypair.pem ubuntu@13.234.239.60
The authenticity of host '13.234.239.60 (13.234.239.60)' can't be established.
ED25519 key fingerprint is: SHA256:Tr/3nOzYeMOXRHX8VbXxMJFNo0OgX8YRqGgYeJE1gkQ
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '13.234.239.60' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sun May  3 08:31:20 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.5% of 6.61GB Processes:    118
Memory usage: 27%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.1.187

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Sun May  3 08:16:01 2026 from 13.233.177.4
ubuntu@ip-10-0-1-187:~$ whoami
ubuntu
```

Accessed the internet to download and install a web server

```

15 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-10-0-1-187:~$ sudo apt install apache2 -y
Installing:
  apache2

Installing dependencies:
  apache2-bin  apache2-data  apache2-utils  libapr1t64  libaprutil1  libaprutil1-dbd-sqlite3  libaprutil1-ldap  libaprutil1t64  liblua5.4-0  ssl-cert

Suggested packages:
  apache2-doc  apache2-suexec-pristine | apache2-suexec-custom  www-browser

Summary:
Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 15
Download size: 2141 kB
Space needed: 8233 kB / 4697 MB available
```